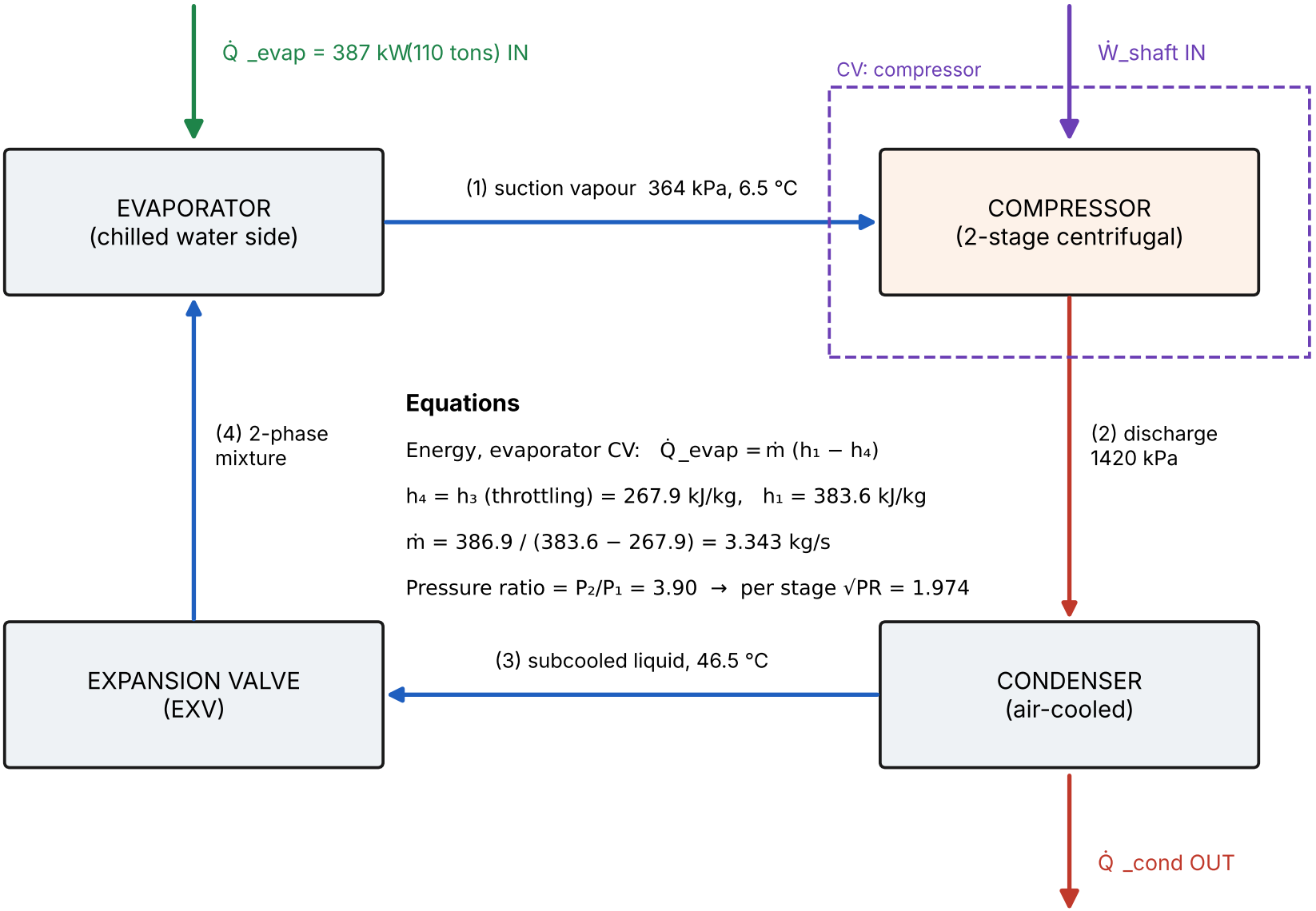
FBD-1

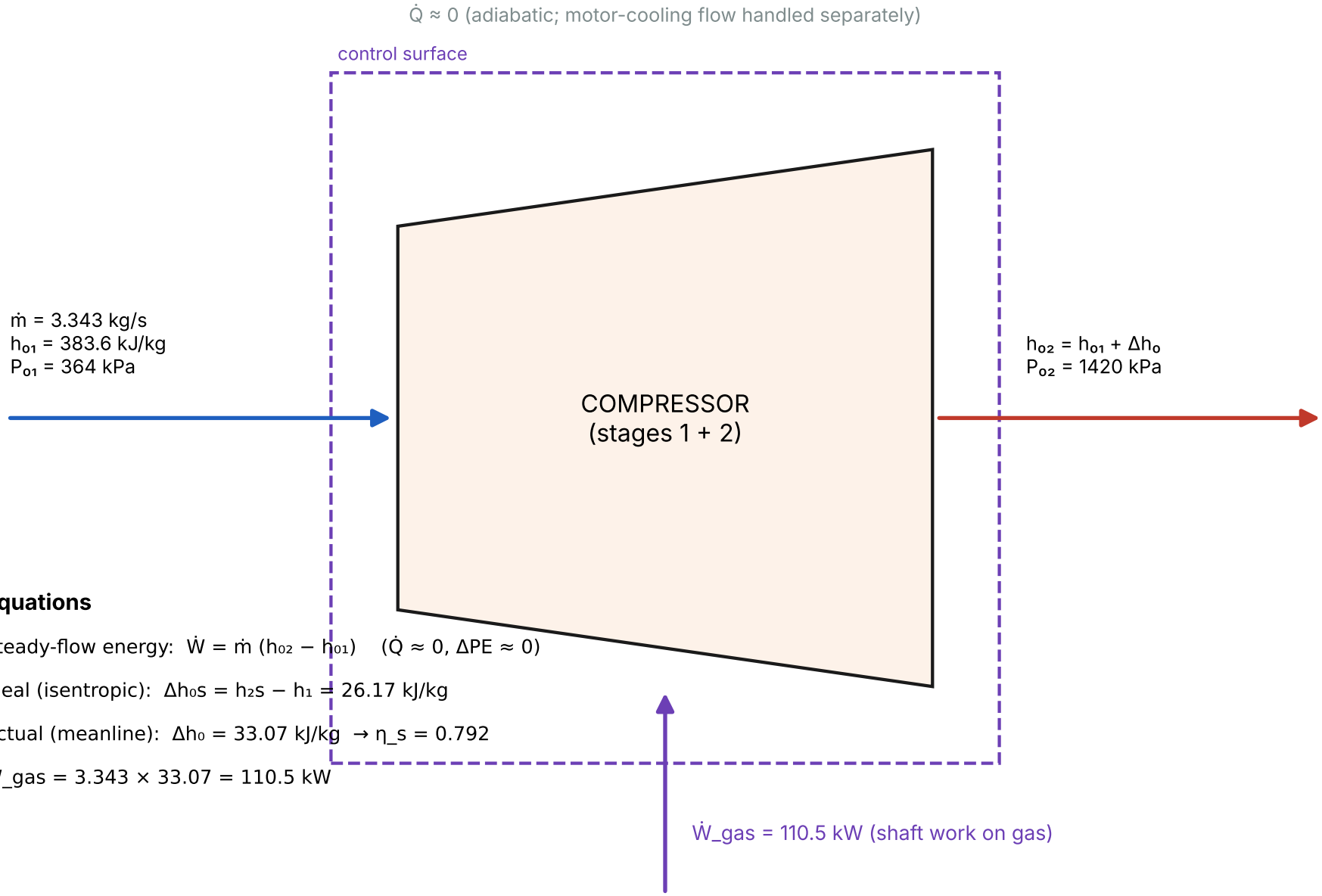
Chiller refrigeration cycle: system control volumes



Turbocor-class compressor study: free-body / control-volume diagrams. Numbers from scripts/hand_calcs.py.

FBD-2

Compressor control volume: first law (steady flow)



Equations

Steady-flow energy: $\dot{W} = \dot{m}\,(h_{02} - h_{01})$ $(\dot{Q} \approx 0,\ \Delta PE \approx 0)$

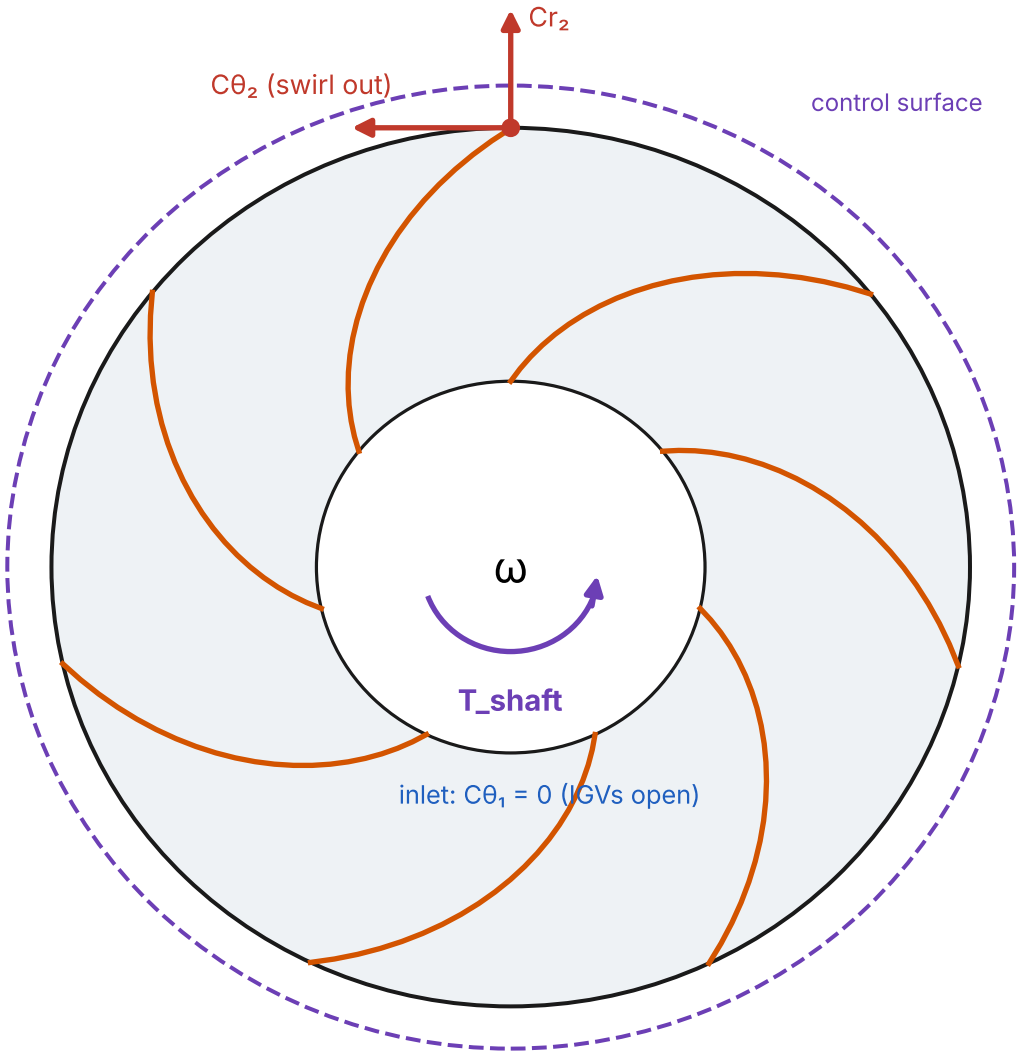
Ideal (isentropic): $\Delta h_{0s} = h_{2s} - h_1 = 26.17$ kJ/kg

Actual (meanline): $\Delta h_0 = 33.07$ kJ/kg → $\eta_s = 0.792$

$\dot{W}_{gas} = 3.343 \times 33.07 = 110.5$ kW

FBD-3

Impeller control volume: angular momentum (Euler turbomachine equation)



Equations

Angular momentum about the shaft axis:

$T_{shaft} = \dot{m}\,(r_2 C\theta_2 - r_1 C\theta_1)$

Power: $\dot{W} = T\,\omega = \dot{m}\,(U_2 C\theta_2 - U_1 C\theta_1)$

With $C\theta_1 = 0$: $\Delta h_0 = U_2\, C\theta_2$

Slip (Wiesner): $\sigma = 1 - \sqrt{(\cos \beta_2 b)} / Z^{0.7}$

$Z = 14$, $\beta_2 b = 40° \rightarrow \sigma = 0.862$

$C\theta_2 = \sigma U_2 - Cm_2 \tan \beta_2 b, \quad Cm_2 = \varphi_2 U_2$

$\psi = \Delta h_0 / U_2^2 = \sigma - \varphi_2 \tan \beta_2 b = 0.627$

$U_2 = 162.4$ m/s $\rightarrow \Delta h_0 = 16.53$ kJ/kg

$D_2 = 60 U_2 / (\pi N) = 103.4$ mm (N = 30,000 rpm)

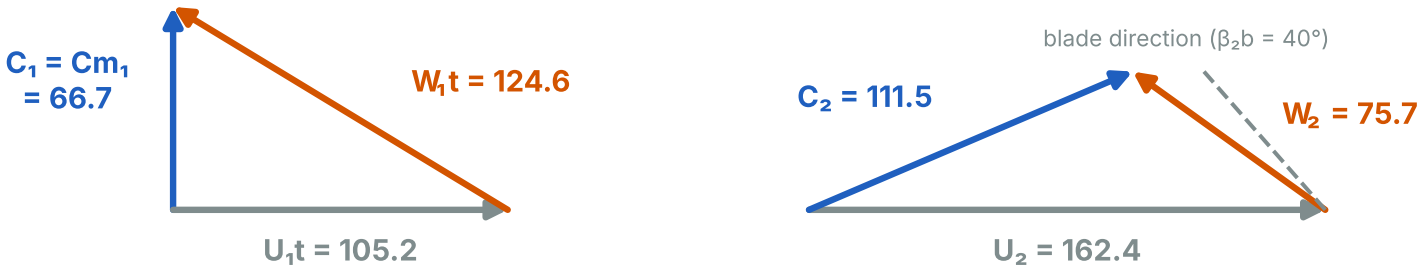
Stage-1 torque $= \dot{m} \Delta h_0 / \omega = 17.6$ N·m

FBD-4

Velocity triangles (kinematics): stage-1 inlet tip and exit, to scale

INLET (blade tip) **EXIT (r_2 = 51.7 mm)**

$\beta_1 t = \text{atan}(U_1 t / Cm_1) = 57.6°$ from axial; $M_1 rel = W_1 t / a_1$ ~~$C\theta_2 88$~~ $\sigma U_2 - Cm_2 \tan\beta_2 b = 101.8$; $\alpha_2 = \text{atan}(C\theta_2 / Cm_2) = 65.9°$ from radial



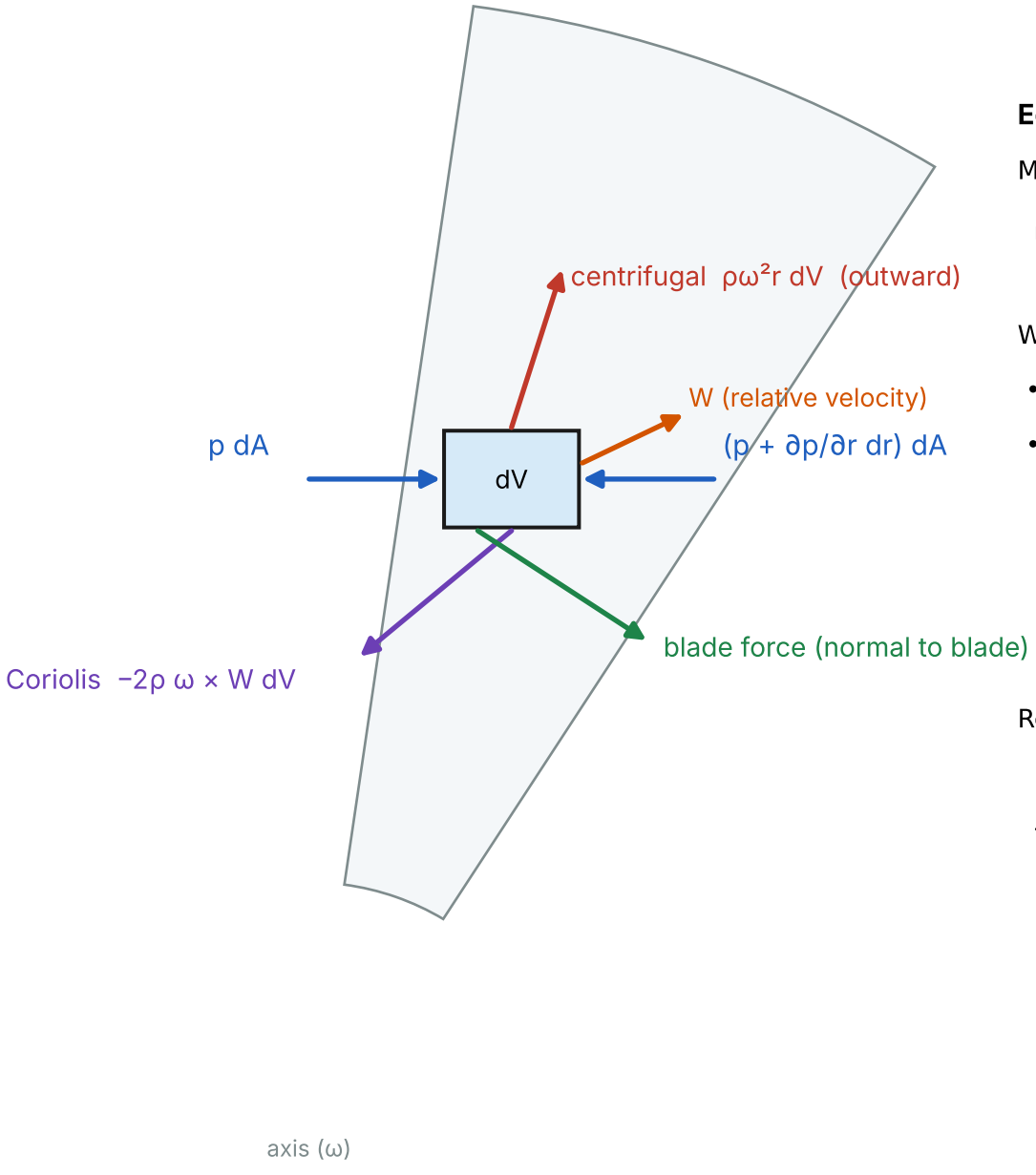
50 m/s

Vector relation at every point: C = U + W (absolute = blade + relative)

Slip: the gas leaves at a smaller swirl than the blade angle implies (σ < 1)

FBD-5

Fluid particle in the rotating passage (relative frame): forces per unit volume



Equations

Momentum, rotating frame (steady):

$$\rho(W\cdot\nabla)W = -\nabla p + \rho\omega^2 r\, e_r - 2\rho\,\omega\times W + \nabla\cdot\tau$$

Why CFX solves in the rotating frame:

- blades and hub are fixed → steady problem
- the frame adds two 'apparent' forces:

  centrifugal $\rho\omega^2 r$ (pumps gas outward)

  Coriolis $-2\rho\omega\times W$ (loads the blades,

  creates the pressure/suction sides)

Rothalpy is conserved along a streamline:

$$I = h + W^2/2 - U^2/2 = \text{const}$$

$$\rightarrow \text{static enthalpy rise} \approx (U_2^2 - U_1^2)/2 + (W_1^2 - W_2^2)/2$$

FBD-6

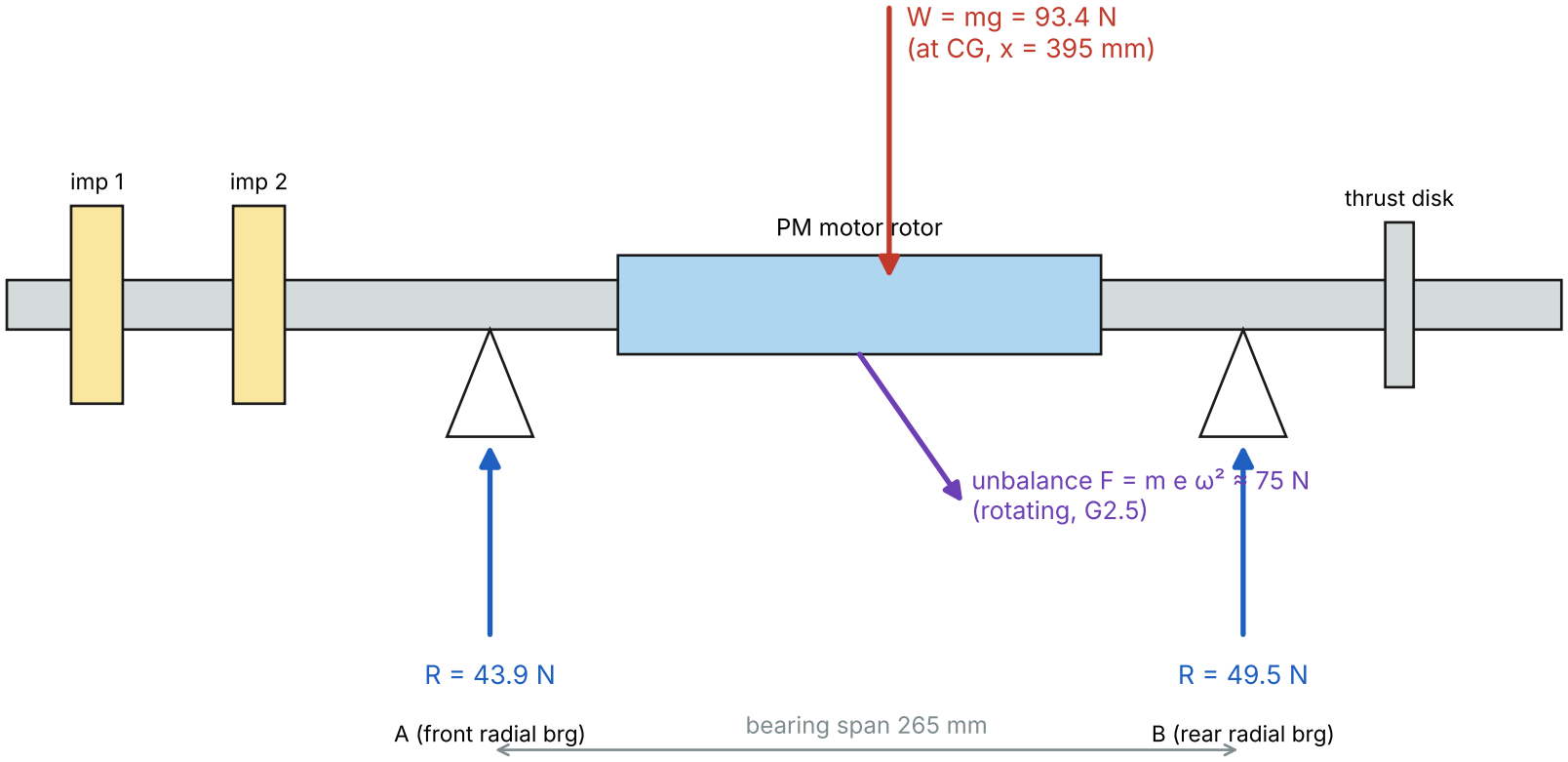
Rotor: radial free-body diagram (gravity, bearings, unbalance, torque)

$\Sigma F_y = 0$: $R_A + R_B = W = 9.52\ \text{kg} \times 9.81 = 93.4\ \text{N}$

$\Sigma M_A = 0$: $R_B (x_B - x_A) = W (x_{CG} - x_A) \rightarrow R_B = 49.5\ \text{N}$, $R_A = 43.9\ \text{N}$

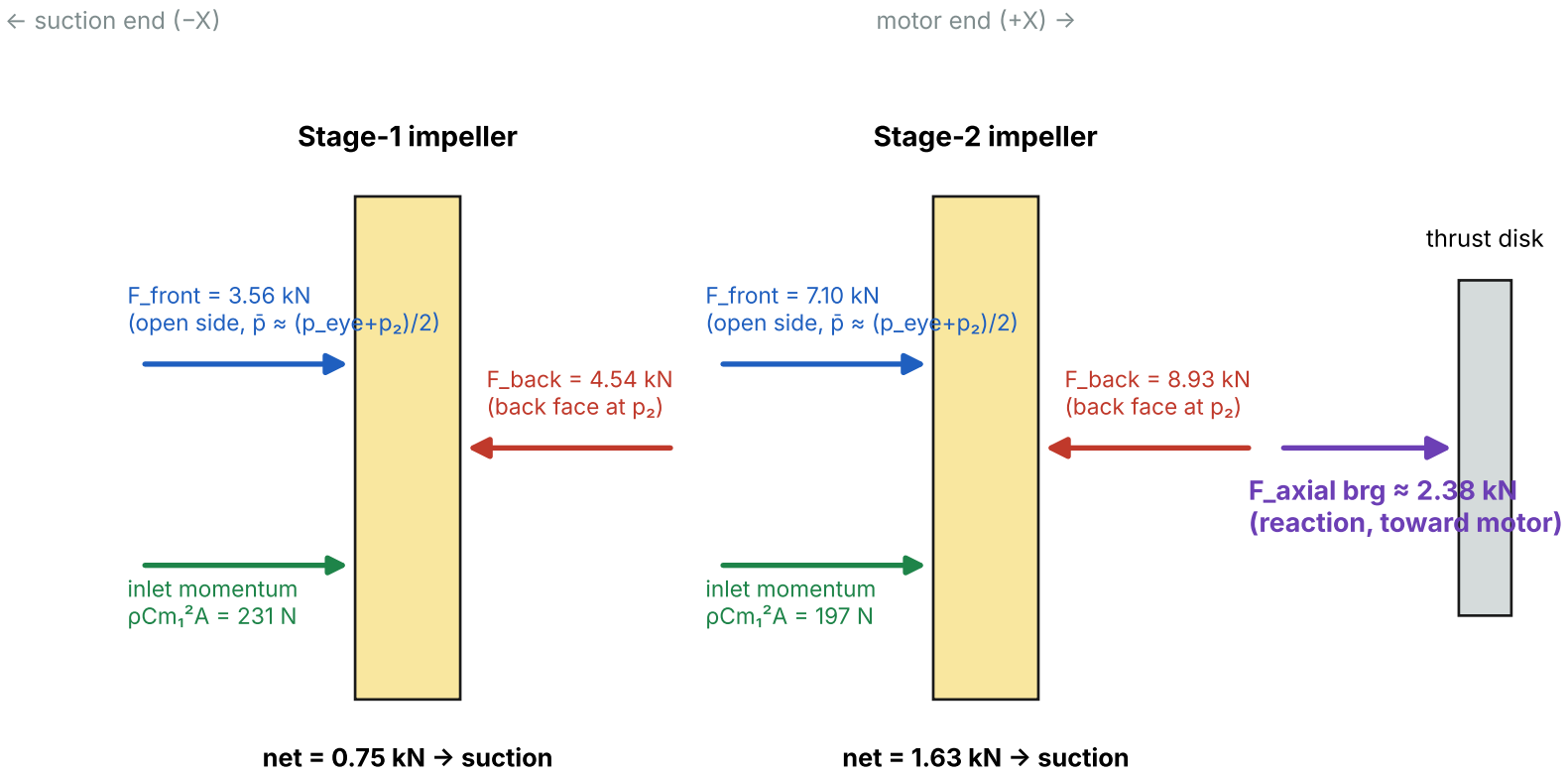
Unbalance (ISO 1940 G2.5): $e = G/\omega = 0.80\ \mu\text{m} \rightarrow F = m\, e\, \omega^2 = 75\ \text{N}$ (rotates at 1×)

Torque balance (steady): $T_{motor} = T_{aero} = 35.9\ \text{N·m}$



FBD-7

Impellers: axial free-body diagram (pressure thrust, momentum, axial bearing)



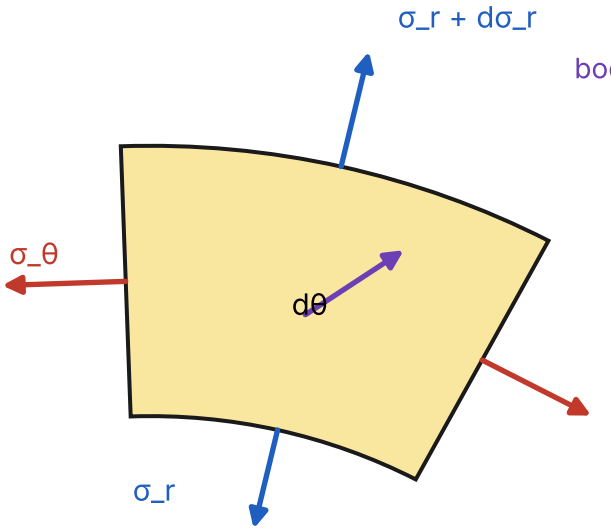
Stage 1: p_eye = 324 kPa, p_2 = 572 kPa; Stage 2: p_eye = 665 kPa, p_2 = 1124 kPa

ΣF_x = 0: F_axial = Σ(F_back − F_front − ρCm₁²A) = 2.38 kN (net push toward suction; ESTIMATE)

Real machines cut this with labyrinth seals / balance holes / back-to-back cavities; the axial bearing takes the rest.

FBD-8

Impeller disc: differential element under centrifugal load (overspeed check)



Equations

Radial equilibrium of the element:

$$\frac{d(r\,\sigma_r)}{dr} - \sigma_\theta + \rho\omega^2 r^2 = 0$$

Solution, flat disc with bore r_b (plane stress):

$$\sigma_{\theta,\max}\ (\text{at bore}) = \frac{3+\nu}{4}\cdot\rho\omega^2\left[r_2^2 + \frac{1-\nu}{3+\nu}\,r_b^2\right]$$

7075-T6: $\rho = 2810$ kg/m³, $\nu = 0.33$, $S_y = 503$ MPa

$r_2 = 51.7$ mm, $r_b = 9$ mm

30,000 rpm: $\sigma_\theta = 62$ MPa

34,500 rpm (115 %): $\sigma_\theta = 82$ MPa → SF = 6.1

39,000 rpm (max): $\sigma_\theta = 105$ MPa

Blades add load at the hub → confirm with FEA.